A Subspace Selection-Based Discriminative Forest Method for Hyperspectral Anomaly Detection

Shizhen Chang, Bo Du, *Senior Member, IEEE*, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*— In this article, a new subspace selection-based discriminative forest (SSDF) method is proposed for the anomaly detection of hyperspectral remote sensing imagery. Most of the existing anomaly detection approaches construct a profile of background instances and then identify instances that do not conform to the background materials as anomalies. However, this type of method generally fails to avoid the background contamination caused by abnormal targets. In this case, we borrow from the concept of isolation and propose an isolation-based discriminative forest model which exploits subsampling rather than modeling the background instances. Furthermore, considering that the data volume of a hyperspectral image is usually huge, the proposed discriminative forest model explores a subspace selection process while splitting the leaf nodes of the binary trees to preserve those bands containing crucial abnormal target information and improve the reliability of the tree-splitting criterion. The proposed detector successfully integrates dimensionality reduction and the data-splitting technique to define pixels as anomaly or background. The extensive experimental results obtained with four hyperspectral data sets demonstrate that the proposed SSDF algorithm outperforms the other state-of-the-art algorithms and hence provides a new perspective for the anomaly detection of hyperspectral remote sensing imagery.**

***Index Terms*— Anomaly detectionaxis-parallel subspace, dimensionality reduction, hyperspectral imagery, isolation-based forest.**

I. Introduction

SINCE hyperspectral images (HSIs) span the visible to infrared spectrum with hundreds of contiguous and narrow spectral bands [1], they can provide us with abundant information about the land cover [2]. Due to the high spectral resolution, HSIs can provide improved target detection and classification capabilities, and thus play an important role in many fields, such as marine remote sensing [3], vegetation research [4], [5], fine agriculture [6], geological surveying [7], atmospheric and environmental remote sensing [8], military reconnaissance [9], camouflage identification [10], etc. Target detection [11], which is aimed at identifying the pixels of interest using prior information of the target spectra or separating those targets that have significant differences with their background, is the focus of our research [12], [13].

According to the availability of prior knowledge of the target spectra, the detection problem can be divided into two categories: 1) signature-based target detection, where we are provided with prior knowledge of the spectral characteristics of the desired targets [14], [15]; and 2) anomaly detection, where no prior knowledge of any target or background is used or assumed [16]. Generally speaking, prior knowledge of the background is difficult to obtain, which leads to a lack of background training information [17]. Moreover, the target subpixel and spectral variability problems also increase the difficulty of separating the specific targets from various backgrounds [13]. These obstacles make anomaly detection a more practical approach. In this article, we focus on solving the anomaly detection problem for hyperspectral remote sensing. Based on binary hypothesis testing theory, each pixel is labeled as background (H_0) or anomaly (H_1) [18]. The detection result is then derived from the likelihood ratio (LR) or the generalized LR (GLR) of the conditional probability distribution functions (pdfs) of the anomaly and background pixels under the two hypotheses [19].

Many different hyperspectral anomaly detection algorithms have been proposed and used in recent years [20]–[23]. Most studies have focused on constructing an appropriate profile of the background instances, where the background is sufficiently suppressed and the anomalies are highlighted [24]. Many of the classical detectors, such as the Reed-Xiaoli (RX) detector [25], the cluster-based anomaly detector (CBAD) [26], blocked adaptive computationally efficient outlier nominators (BACONs) [27], and the random selection-based anomaly detector (RSAD) [28], focus on estimating the background statistical distribution. The RX algorithm, which is viewed as the benchmark of the anomaly detection methods, uses the Mahalanobis distance between the test pixel and its neighboring background pixels to measure the abnormality. The CBAD method uses a clustering algorithm to estimate the component mean and covariance matrices and to segment the image, and it then determines a pixel's class membership based on the minimum Euclidean distance measurement. The BACON method applies an iterative background estimation strategy to avoid anomaly contamination in the background estimation and uses the Mahalanobis distance to compute the

Manuscript received April 8, 2019; revised August 9, 2019 and November 18, 2019; accepted November 21, 2019. Date of publication January 15, 2020; date of current version May 21, 2020. This work was supported in part by the National Natural Science Foundation of China under Grant 41431175 and Grant 41871243 and in part by the Science and Technology Major Project of Hubei Province (Next-Generation AI Technologies) under Grant 2019AEA170. *(Corresponding author: Liangpei Zhang.)*

Shizhen Chang and Liangpei Zhang are with the Remote Sensing Group, State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Institute of Artificial Intelligence, Wuhan University, Wuhan 430079, China (e-mail: szchang@ whu.edu.cn; zlp62@whu.edu.cn).

Bo Du is with the National Engineering Research Center for Multimedia Software, Institute of Artificial Intelligence, School of Computer Science, Wuhan University, Wuhan 430072, China (e-mail: remoteking@whu.edu.cn).

Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2019.2960391

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:17:50 UTC from IEEE Xplore. Restrictions apply.

degree of the anomaly of the test samples in each iteration. The RSAD method randomly selects pixels from the image to contribute to the background statistics and utilizes a number of random selections and a fusion operation to reduce the potential contamination caused by anomaly pixels.

The aforementioned methods assume that the hyperspectral data obey a Gaussian distribution or a multivariate Gaussian distribution, and obtain the detection result by establishing a hyperellipsoid with the estimated mean vector and the covariance matrix [29]. However, the anomalies and the backgrounds in real-world HSIs usually have higher-order relationships, and the Gaussian/multivariate Gaussian assumptions in these algorithms under the two hypotheses are generally not valid. Therefore, many algorithms formulate nonlinear versions by mapping the original data space into a high-dimensional feature space to produce a nearly Gaussian distribution. Examples of such algorithms are kernel RX (KRX) [30], the selective kernel principal component analysis (KPCA) algorithm [31], the robust nonlinear anomaly detector (RNAD) [32], the support vector data description (SVDD) anomaly detector [33], etc. KRX, which simultaneously takes the nonlinear characteristic and high-order relationship of the spectral signatures into consideration, projects the hyperspectral data to a high-dimensional kernel Hilbert data space and provides a nonlinear version of the RX algorithm for hyperspectral anomaly detection. The selective KPCA algorithm extracts the anomalous information to the maximum extent and suppresses the background clutter in the selected component. The RNAD method combines the kernelization procedure with robust regression analysis to formulate a new nonlinear robust detector which can effectively avoid the contamination of the Mahalanobis distance-based background features caused by anomalies in a high-dimensional feature space. The SVDD algorithm utilizes a single-class support vector machine (SVM) classifier to estimate the minimum enclosing hypersphere that contains most normal instances for a given data set, where pixels that fall outside of the hypersphere are considered as anomalies. The SVDD classifier finds the radius and the center of the enclosing hypersphere in a high-dimensional kernel feature space and does not require any prior assumption about the data distribution or training samples.

Instead of estimating a statistical model and kernel feature for the background, many geometrical modeling-based methods have been proposed, which assume that the background can be approximately represented by a group of major spectra or bases extracted from the image. Typical examples are the subspace-based methods, such as the background joint sparse representation detector (BJSRD) [34], the collaborative representation-based detector (CRD) [35], the low-rank and sparse representation (LRASR) anomaly detection method [36], and the tensor decomposition-based anomaly detector [37]. The BJSRD algorithm assumes that the background can be modeled as lying in a low-dimensional subspace and the CRD algorithm uses the role of collaboration to trace which kind of landcover the test sample belongs to, both of their background dictionaries are constructed by a set of bases that are acquired from a dual local window. They then project every pixel to the background orthogonal subspace and calculate the representation residual of each pixel to determine its abnormality. The LRASR detector is based on the separation of the anomaly and the background parts, where the background is assumed to lie in the lowest-rank representation of the image. This method applies the LRASR technique to solve the background subspace problem. The tensor decomposition-based anomaly detection method utilizes the third-order tensor representation to describe and exhibit the spectral and spatial characteristics of the anomalies.

The aforementioned methods are all model-based anomaly detectors that construct a background distribution model and identify pixels that do not obey the background distribution as anomalies. The two major drawbacks of these approaches are: 1) the detectors are optimized to profile the background statistical distribution, but they are not optimized to detect anomalies. As a consequence, their results might not be as good as expected. They also cannot avoid the background contamination caused by abnormal pixels, which can result in too many false alarms or too few anomalies being detected; 2) hyperspectral data sets contain a large amount of redundancy, and thus some undesirable statistical and geometrical properties. Moreover, this drawback makes the dimensionality reduction techniques more attractive to represent the main information of the data, without losing effectiveness.

Different from previous works, the proposed subspace selection-based discriminative forest (SSDF) method uses a different type of isolation-based method to detect anomalies. The proposed method brings a new perspective for solving anomaly detection with no need for background assumption. The outliers can be separated by means of repeated learning the values of subsamples. Since the anomalies are usually rare and different in the image scene, they also tend to be more easily isolated from the background pixels by appropriate splitting criterion. The proposed SSDF method has two main stages: an ensemble stage and a scoring stage. The first stage involves building a forest of discriminative binary trees using subsamples of the image and applying an axis-parallel subspace selection process to preserve the original structure of the high-dimensional data space. The second stage involves testing every pixel of the image through the discriminative forest model to obtain the abnormality score as the detection output. The novelties and contributions of the proposed SSDF method can be summarized as follows.

1) The isolation-based method enables the discriminative trees to approximate the overall data distribution by estimating partial models that concentrate on separating the isolated points from the background pixels. This method takes full account of the fact that the background is concentrated within a certain range and anomalies are sparse and discrete, which reduces the swamping and masking effects during the construction of the background model.
2) The axis-parallel subspace selection process is applied to reduce the dimensionality of each discriminative tree, which can reasonably retain the information of the original space as well as reduce the spectral similarity between the bands of hyperspectral data. This manipulation helps us to avoid the band redundancy in HSIs

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:17:50 UTC from IEEE Xplore. Restrictions apply.

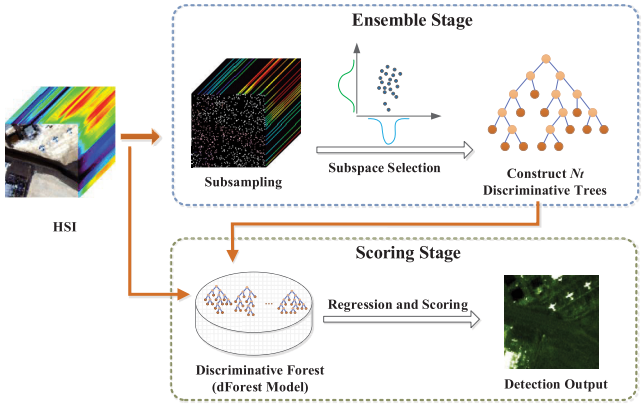


Fig. 1. Flowchart of the proposed SSDF method for HSI anomaly detection.

and improves the discriminant reliability at the tree leaf node.

3) Different from previous background modeling detectors, subsample profiling is used to simulate and estimate the overall data distribution, which allows the SSDF method to effectively detect anomalies in extremely large data sets. The dimensionality reduction and data-splitting technique are integrated to avoid high-dimensional problems with unreliable splitting decisions.

The rest of this article is organized as follows. Section II presents the derivation of the SSDF method. Section III describes the extensive experimental results and presents a sensitivity analysis for the parameters of the proposed algorithm. The conclusion of this article is given in Section IV.

II. Subspace Selection-Based Discriminative Forest Method

The proposed SSDF algorithm models the data set through a forest of binary discriminative trees, based on the concept of "isolation," while using a subspace selection method at each leaf node to overcome the dimensional redundancy of HSIs. The basic process flow of the SSDF method is illustrated in Fig. 1, and the detailed description is shown in the following order: the axis-parallel-based subspace selection process, which is used for improving the stability of discriminative trees, is introduced in Section II-A. Section II-B gives a definition of the proposed isolation-based discriminative forest model for readers. The two stages of the proposed SSDF method, namely the ensemble stage and scoring stage, are finally described in Section II-C.

A. Axis-Parallel Subspace Selection Algorithm

Many of the existing anomaly detection methods [38]–[40] in data-mining applications rely on computing a full-dimensional distance, in order to examine the abnormality of each object. However, objects are almost all equidistant in the high-dimensional space, which results in the distance-based methods failing to differentiate anomalies from the normal instances [41]. This challenge arising from the "curse of dimensionality" is not specific to anomaly detection, and it also affects many problems such as clustering and similarity search. Moreover, some studies have found that the outlier-generating mechanisms are only relevant to a subset of features, i.e., the subspaces of the original data space [42]. In view of this assumption, the subspace-based methods are generally applied to solve the full-dimensional anomaly detection problem. Among the different methods, the axis-parallel subspace method has been proven to be an effective feature selection method that can reduce these irrelevant attributes [43].

In this article, the theory of the axis-parallel subspace selection process is applied to the ensemble stage of the proposed SSDF method to help choose those relevant attributes for tree splitting. Its general idea is selecting a subset of bands as discriminative attributes and analyzing how well each object fits the subspace by other reference points. The subspace spanned by a set of points is simply an axis-parallel hyperplane of any dimensionality $l < L$, where L is the dimensionality of the entire feature space, such that all points of the reference set are close to this hyperplane. If a point deviates significantly from this reference hyperplane, it is considered to be an outlier in the subspace that is perpendicular to that hyperplane.

For a given hyperspectral data set $X = [x_1, \ldots, x_N]$, $x_i \in R^L, i = 1, \ldots, N$, where N is the number of pixels and L is the total number of bands, dist represents the Euclidean distance function between pairwise points. For any point $x_i \in R^L$, $i = 1, \ldots, N$, we denote the projection of x_i onto the jth band by x_{ij}. The mean variance is used as the judgment of the dispersion of X. For a dimension j, its mean variance can be represented as

$$\mathrm{var}_j = \frac{\sum_{i=1}^{N} (\mathrm{dist}(x_{ij} - \mu_j))^2}{\mathrm{Card}(X)} \tag{1}$$

where $\mathrm{Card}(X)$ denotes the cardinality of X. We learned from the axis-parallel subspace selection algorithm that the distribution of the data on the bands unrelated to the discrimination of abnormal information is more dispersed, and therefore the variance var_j is larger. Furthermore, since the distribution of the data on the dimensions related to the discrimination of abnormal information is more concentrated, the variance is relatively smaller. As a result, the axis-parallel subspace selection method differentiate the variance of each dimension from low to high, and select those bands with a smaller var_j as the basis of computing splitting criterion for the proposed SSDF model.

B. Isolation-Based Discriminative Forest Model

The term ***isolation*** was first proposed by Liu *et al.* [44] at the Eighth IEEE International Conference on Data Mining, and means "separating one object from the rest of the objects." Liu *et al.* [44] designed a collection of binary trees to randomly partition instances until all of them are isolated, since the anomalies which participate in a few propagations of the image are apt to have shorter paths in the binary tree, and the anomalies, which usually have more distinguishable attribute values, can be more easily separated [45]. This random partitioning process results in the anomalies having shorter path lengths than the normal instances. The aforementioned binary trees are named as isolation trees (iTrees),

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:17:50 UTC from IEEE Xplore. Restrictions apply.

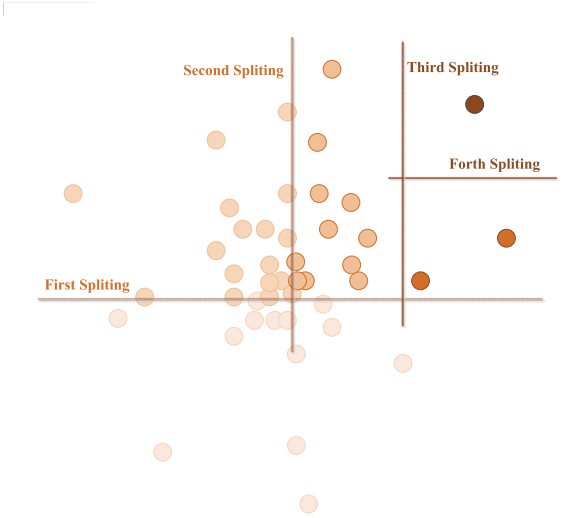


Fig. 2. Arbitrarily splitting iTree with a group of data points in a 2-D space.

and Fig. 2 shows an arbitrarily splitting iTree with a group of data points in a 2-D space. We can clearly see that the discrete points can be isolated much more easily than the scattered points, which results in them having a shallower depth on the iTree. Therefore, when a forest of iTrees (an iForest) collectively produces shorter path lengths for some points, these points are more likely to be anomalies [46]. By a repeated collection of subsamples, the iForest model reduces the swamping and masking effects. As a result, the iForest model can show state-of-the-art performance in the data-mining field, with low linear time complexity and a low memory requirement.

Returning to the anomaly detection problem in HSIs, the concept of isolation is also applicable. Since the anomalies are usually "rare and different" in the image scene, they also tend to be more easily isolated from the background pixels by appropriate splitting criterion. The aforementioned binary classification trees can repeatedly split the data into two classes at each leaf node until all the points are isolated and the path length can be utilized as the judgment to describe how abnormal an object is.

Since the original iForest model randomly selects one attribute from the objects of the current node and segments the objects by the values of the selected attribute as the segmentation standard, the number of split meshes grows exponentially as the data dimension increases, and its splitting reliability is greatly reduced when the values are strongly correlated. Therefore, based on the concept of "isolation," we proposed an improved SSDF method which is more applicable for the hyperspectral anomaly detection problem. This section provides formal definitions and descriptions of the discriminative tree and the discriminative forest.

Definition 1 *(Discriminative Tree):* We let T be the discriminative binary tree, and t be the current leaf node of T. t is either defined as an external node that has no child, or an internal node with one test that has two child nodes (t_l, t_r), where t_l and t_r represent the left child node and the right child node of t, respectively. The test, for which we know the specific subspace and the splitting threshold t_s, is aimed at dividing the included data in the current node into two classes, based on the splitting criterion computed by the selected bands. The pixels having smaller values than t_s are allocated into t_l, and those having values larger than t_s are allocated into t_r.

We consider a given subsample set $X_n = [x_1, \ldots, x_n]$ of X, where n is the number of pixels to build the discriminative tree and L is the number of bands of this data. Based on the concept of "isolation", X_n is recursively divided by computing the splitting criterion by the selected subspace and determining the splitting threshold t_s by computing the Gini index of the splitting criterion, until either: 1) the tree reaches the height limit; 2) the current node only has one pixel, i.e., $|X_n| = 1$; or 3) all the data in X_n have the same values. The discriminative tree is a tough binary tree, where each node in the tree has exactly zero or two child nodes. Hence, if all the pixels are nonoverlapped, the structure of the discriminative tree with n elements is constructed to a full binary tree with n external nodes and $n-1$ internal nodes.

Definition 2*:* The **Path Length** $e(x_i)$ expresses the number of edges x_i traversed in a discriminative tree from the root node to the external node. The task of anomaly detection is to obtain a result that can distinguish the anomalies from the overall data set. Taking into account that an anomaly is more likely to be an external node, it usually has a shorter path from the external node to the root node. Thus, the path length of a point x_i can be adopted as the basis of the anomaly score to reflect its degree of abnormality.

Considering that the discriminative tree model has a similar structure to the binary search tree (BST) model, the estimation of the path length is inspired by the BST search analysis

$$p(n) = 2H(n-1) - (2(n-1)/n) \tag{2}$$

where $p(n)$ is the average height of the tree; n is the total tree elements of the discriminative tree; and $H(\cdot)$ is the harmonic number, which is approximately equal to $\ln(\cdot)+0.5772156649$ (Euler's constant).

Therefore, we finally derive the anomaly score of pixel x_i after normalizing the path length

$$S = 2^{-\frac{E(e(x_i))}{p(n)}} \tag{3}$$

where $E(e(x_i))$ is the average of $e(x_i)$ from a collection of discriminative trees.

C. Subspace Selection Based Discriminative Forest Method

In the ensemble stage, the SSDF method constructs a forest of discriminative trees by subsamples which are randomly selected from the image. The discriminative tree is a full binary tree whose nodes are either external without any child or internal with both left child and right child. Considering the serious band redundancy phenomenon that exists in HSIs and the strong spectral similarity between different bands, the discriminative tree first selects a number of bands from the current data set based on the aforementioned axis-parallel subspace selection process, and then uses the selected bands to compute the splitting criterion of the subsequent leaf node.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:17:50 UTC from IEEE Xplore. Restrictions apply.

If we assume that l bands are selected as the reference space, then the splitting criterion of the pixel x_i can be represented as

$$c(x_i) = \sqrt{\sum_{j=1}^{l} (\mathrm{var}_j(x_i) \cdot (\mathrm{dist}(x_{ij} - \mu_j))^2)} \tag{4}$$

where $c(x_i)$ represents the splitting criterion of pixel x_i, and $\mathrm{var}_j(x_i)$ is the variance of pixel x_i in the jth band.

As we all know, there are various rules that can be utilized to decide the splitting threshold at each leaf node which is crucial owing to its great impact on the splitting accuracy. There are lots of common criteria that have been proposed to decide an appropriate splitting threshold, such as the information gain criterion [47], the Gini index [48], and Bayesian classification error [49]. In the proposed method, statistical-based information related to the variance of the selected bands is applied as the splitting threshold. According to the splitting criterion, we classify the elements of current tree note t into k classes, and define $p(\cdot \mid t)$ as the probability of the ith class. The estimated probability of the object actually being in class i is then $p(t_i \mid t)$. Therefore, the estimated probability of misclassification under this rule is

$$\begin{aligned} \mathrm{Gini}(t) &= \sum_{i=1}^{k} \sum_{j=1, j\neq i}^{k} p(t_i \mid t) p(t_j \mid t) \\ &= 1 - \sum_{i=1}^{k} ((t_i \mid t))^2. \end{aligned} \tag{5}$$

When used for qualifying splitting, the Gini splitting threshold is decided by minimizing the following expression:

$$\mathrm{GiniSplit}(t) = \frac{|t_l|}{|t|}\mathrm{Gini}(t_l) + \frac{|t_r|}{|t|}\mathrm{Gini}(t_r) \tag{6}$$

where t_l and t_r represent two parts of t, respectively.

We traverse all the instances of the current tree by the splitting criterion from selected subspace, and then splitting these points by means of the Gini splitting threshold until the limitations are reached. In our experiments, the class number is set to $k = 5$ by experience.

While constructing the discriminative tree, the depth of the tree is greatly influenced by the subsample size. If the depth is limitless, a discriminative tree with n elements can have a maximum depth of n, which will waste unnecessary computation time and storage space. On the other hand, we are more concerned about the earlier isolated objects than the leaf nodes at the very end of the tree (those backgrounds with longer path lengths). Therefore, in order to save unnecessary computation and to make a stronger path length contrast between the background and the anomalies, we set the tree height limit to $[\log_2 n]$. The basic procedure of constructing a discriminative tree is described in Algorithm 1, where *dTree* is used to represent "discriminative tree."

After constructing a certain number of binary trees, the discriminative forest is completed by gathering all the trees together. There are two main parameters in the ensemble stage, which are the number of subsamples n selected for each tree and the number of trees N_t, respectively. The sensitivity of the SSDF method to each parameter will be analyzed in Section IV.

Algorithm 1 $dTree$ Construction

Inputs: X-the input HSI image, n-the number of elements in each tree, H-the height of the tree, h_l-the height limit of the tree, and l-the number of bands selected for computing the splitting criterion

Output: a $dTree(X_n, 0, h_l)$

1: Randomly select n pixels X_n from X as the total tree elements of $dTree$
2: **Repeat**
3: Compute the mean variance of each band j for the current tree elements X_n
$\mathrm{var}_j = \sum_{i=1}^{n} (dist(x_{ij} - \mu_j))^2 / Card(X_n)$
4: Selected l bands with smaller var_j and compute the splitting criterion of each pixel x_i:
$c(x_i) = \sqrt{\sum_{j=1}^{l} \left(\mathrm{var}_j(x_i) \cdot \left(dist(x_{ij} - \mu_j)\right)^2\right)}$
5: Decide the splitting threshold t_s based on c (the GINI index)
6: Compare the values of $c(x_i)$ to t_s
7: **Return**
8: **Until** $H > h_l$ or $X_n \leq 1$

In the scoring stage, the detection result of the SSDF method is computed by counting the expected path length of each pixel. For a given test pixel x_i, its path length $e(x_i)$ is undertaken by passing it through each discriminative tree in the forest, as shown in Algorithm 2, and the expected path length $E(e(x_i))$ can be derived by computing the mean of $e(x_i)$s from the discriminative forest.

Algorithm 2 Path Length of x_i

Input: x_i-the test sample, $dTree(X_n, 0, h_l)$-a $dTree$, e-current path length

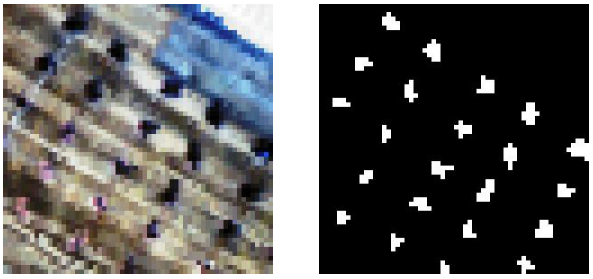
Output: path length of x_i

1: **if** x_i is an external node **then**
2: return $e + c(Card(X_n))$
3: **end if**
4: **if** $c(x_i) < dTree.GiniSplit$ **then**
5: $(dTree.leftnode \leftarrow x_i, e + 1)$
6: **else**$c(x_i) > dTree.GiniSplit$
7: $(dTree.rightnode \leftarrow x_i, e + 1)$
8: **return** result

Then we compute the anomaly score as the detection output of x_i, as shown in (3). When $E(e(x_i))$ is close to 0, the anomaly score S is going to be 1. This means that there are barely no edges that x_i has traversed, and x_i is more likely to be an abnormal pixel. On the other hand, when $E(e(x_i))$ is close to $n-1$, which means that x_i is quite difficult to separate from the data set, it is assumed to be a stable background and the anomaly score S is close to 0.

We finally give an analysis about the time complexity of the proposed SSDF method. As shown in Algorithm 1, there are two main primary operations at the node of *dTree* construction: selecting axis-parallel subspace (line 3) and computing the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:17:50 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 3. (a) San Diego image scene in true color. (b) Ground-truth.

splitting criterion (line 4). For axis-parallel subspace processing, the mean variance needs to be calculated pixelwise, so the worst-case running time is $O(n)$. For the splitting criterion, we need to perform a pixelwise and a bandwise computation of the current tree node, so the worst-case running time is $O(\ln)$. As has been mentioned before, in order to save unnecessary computation and to make a stronger path length contrast between two classes, the tree height limit is set to $\log n$. Therefore, the worst time complexity of the ensemble stage is $O(\ln\log n)$ for a *dTree* and $O(N_t \ln\log n)$ for a *dForest*.

III. Experiments and Analysis

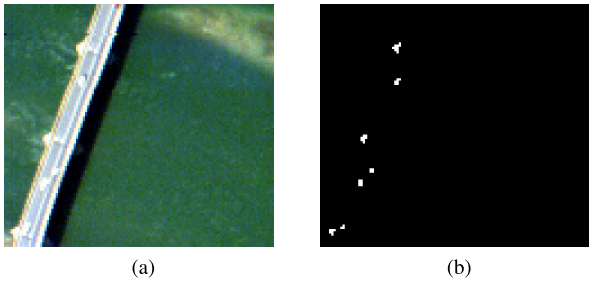
In order to illustrate the effectiveness of the proposed SSDF method, experiments were conducted on four publicly available real-world HSIs, with seven traditional anomaly detection methods as a comparison.

A. Hyperspectral Data Sets

The first data set is a real-world data set collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) from San Diego, CA, USA. The spatial resolution of this image is 3.5 m per pixel. The image has 224 spectral channels in wavelengths ranging from 370 to 2510 nm. After removing the corresponding water absorption regions and low signal-to-noise ratio (SNR) bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224), 189 bands remained for the experiments. The size of the image is 60×60 pixels, and 22 planes (covering 214 pixels) were chosen as the anomalies to be detected in the scene. The true-color image of the San Diego data set and the corresponding ground truth are shown in Fig. 3.

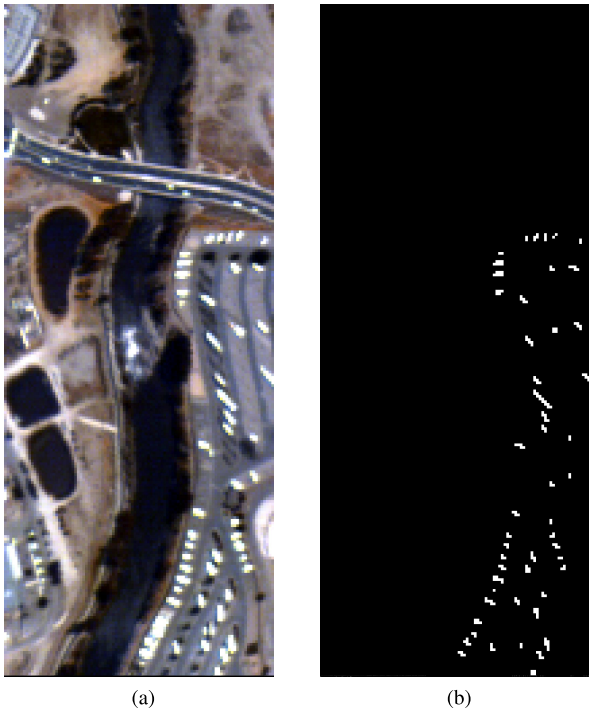
The second data set was collected by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor and is a real-world data set of the city center of Pavia in northern Italy. The spatial resolution of this image is 1.3 m per pixel. The image scene covers an area of 108×120 pixels, with 102 spectral bands in wavelengths ranging from 430 to 860 nm. The main background materials are bridge and water. There are some vehicles on the bridge and bare soil near the bridge pier, represented by 43 pixels in total, which were selected as the anomalies to be detected. The true-color image and its ground truth are shown in Fig. 4.

The third data set covers a scene of Reno, Nevada, USA, and has 356 spectral bands [50]. The spatial resolution of this image is less than 2 m per pixel. This hyperspectral data



(a) (b)

Fig. 4. (a) Pavia image scene in true color. (b) Ground-truth.



(a) (b)

Fig. 5. (a) Reno image scene in true color. (b) Ground-truth.

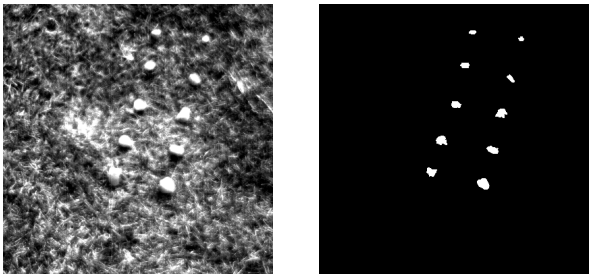
set can be downloaded from http://www.spectir.com/free-data-samples/, and a region of 250×100 pixels was chosen as the test data for our experiments. The anomaly targets to be detected are located on the right side of the scene, which is represented by 214 pixels. The true-color image and the ground truth are shown in Fig. 5.

The fourth hyperspectral data set was acquired by the CRi Nuance hyperspectral sensor [51]. This image covers an area of 400×400 pixels and has 46 spectral channels from 650 to 1100 nm. The spectral resolution of this data set is 10 nm. [52]. The background land-cover types in this image are various kinds of grasses. Meanwhile, there are ten rocks located in the grassy scene, as shown in Fig. 6(a), which, in total, consists of 1254 pixels were selected as the anomalies to be detected, as shown in Fig. 6(b).

B. Detection Performance and Parameter Analysis

1) Evaluation Criteria: In order to analyze and compare the detection efficiency of the different anomaly detectors, three valid evaluation criteria are adopted to qualitatively and quantitatively evaluate the detection performances: the receiver

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:17:50 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 6. (a) Nuance image scene in gray color. (b) Ground-truth.

operating characteristic (ROC) curves [53], the area under the ROC curve (AUC) values, and separability maps between the target and background [54].

When evaluating the performance of given detectors, there is always a compromise between choosing a low threshold to increase the probability of detection P_D and a high threshold to keep the false alarm rate P_{FA} low. The ROC curve is a classic comparison measurement that describes the trade-off between P_{D} and P_{FA} under Neyman-person criterion. In general, the performance of the detector is considered better if having a higher P_D under the same value of P_{FA}.

In addition, the AUC value, which can be considered as an integration of the ROC curve, is derived by calculating the area under the ROC curve. The detector that has a larger AUC value represents a better detection result.

Furthermore, the separability range is also applied to qualitatively evaluate how anomalies are separated from backgrounds by the detector [55]. A better detector usually distinguishes the abnormal targets by highlighting the anomaly pixels and suppressing the backgrounds into small values. Thus, the target-background separability map can be used to evaluate target visibility enhancement. Since the detection results of different methods vary widely, a normalized detection result is adopted. The normalization processing is defined as [56]

$$\hat{D}(\mathrm{x}) = \frac{D(\mathrm{x}) - D_{\min}(\mathrm{x})}{D_{\max}(\mathrm{x}) - D_{\min}(\mathrm{x})} \tag{7}$$

where $D(\mathrm{x})$ is the detection result, $D_{\min}(\mathrm{x})$ and $D_{\max}(\mathrm{x})$ are the minimum and the maximum values of all detection results, and $\hat{D}(\mathrm{x})$ is the final detection result of the data point x after normalization processing. Based on the ground truth, these normalized values are separated into two classes and represented by box charts. We use red boxes to illustrate the distribution of the anomalies and blue boxes to illustrate the background. The box of each class encloses 10% to 90% of the statistical distribution values, and the range between the top and bottom lines contains the 1% to 99% statistical distribution values of each column.

2) Detection Performance: In the experiments, the proposed SSDF method was compared to seven classic anomaly detectors: RX [25], CBAD [26], BACON [27], KRX [30], low probability anomaly detection (LPAD) [57], CRD [35], and the sparsity score estimation framework for hyperspectral anomaly detection (SSEAD) [58]. These methods are either frequently cited in the literature or have shown excellent performances on real hyperspectral data sets. All of the compared algorithms were implemented with their optimal parameter settings.

For the San Diego hyperspectral data set, the 2-D color detection maps of the eight methods are shown in Fig. 7. It can be seen from Fig. 7 that the proposed SSDF method clearly highlights the anomalies from the background land cover. Comparing the detection details to those of the other methods, the suppression effect of SSDF on noise is superior to that of RX, CBAD, and LPAD. The robust iterative background processing makes the results of the BACON algorithm become oversuppressed, which results in an overall darker detection map for the San Diego data set. In contrast, the kernel computation of KRX maps the original image into a higher dimensional feature space. Considering that the discrimination of the land covers is greater in this feature space, the detection values of KRX in the whole image eventually become higher. In comparison, CRD and SSEAD achieve satisfactory background suppression results, while CRD has a slightly higher FAR, and SSEAD appears to be weak in extruding anomalies. Fig. 8 gives a quantitative comparison of the different anomaly detectors with ROC curves and normalized background-anomaly separation maps for the San Diego data set. From Fig. 8(a), it can be observed that for lower FARs, the proposed SSDF method obtains higher probabilities of detection than the other algorithms (e.g., when the FAR is 0.001–0.01). From Fig. 8(b), it can be found that all of the compared algorithms are overlapped over the majority parts (1%–99%) of the background and anomaly boxes. Due to the remarkable anomaly separation effect of the SSDF algorithm, the intersection range between the background box and anomaly box is the smallest. Therefore, the proposed SSDF method achieves a better background-anomaly separation than the other methods.

For the Pavia hyperspectral data set, the 2-D color detection maps of the eight methods are shown in Fig. 9. It can be seen that the RX, CBAD, BACON, and LPAD methods obtain similar visualization results on this image. The output values of the background pixels are relatively small, so most of the anomaly pixels on the bridge are extruded. KRX, CRD, SSEAD, and the proposed SSDF method can better extract anomaly targets than the aforementioned methods. However, their suppression effect on the edges of the bridge seems to be invalid, which can amplify the false alarms. Fig. 10 gives a quantitative comparison of the different anomaly detectors with ROC curves and background-anomaly separation maps for the Pavia data set. We can see that although the visual interpretation results vary significantly, the ROC curves of the RX, CBAD, LPAD, and SSEAD algorithms are very close and cross each other, which makes it difficult to distinguish their detection effects. Meanwhile, BACON and KRX achieve better ROC curves than the other methods, and they can improve the probability of detection at low FARs. The proposed SSDF method obtains the best ROC curve. When the FAR is 0.001, the probability of detection for SSDF is higher than 0.95. At the same time, the probability of detection quickly reaches 100% before the FAR increases to 0.01. It can be seen from Fig. 10(b) that all of the algorithms can separate the middle 80% of the two classes, in which RX, CBAD,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:17:50 UTC from IEEE Xplore. Restrictions apply.

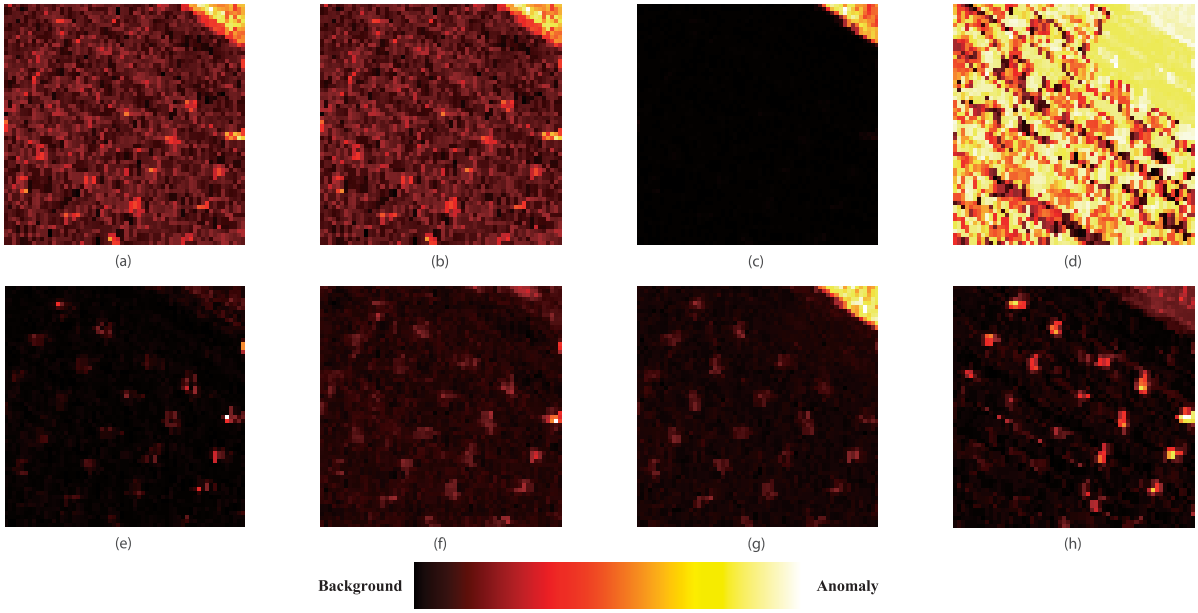
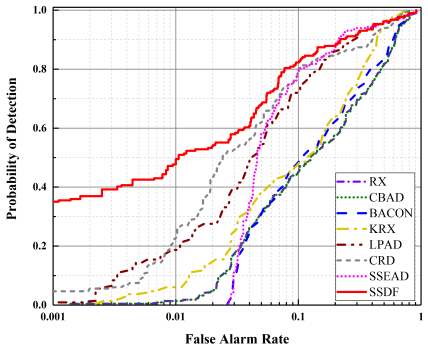
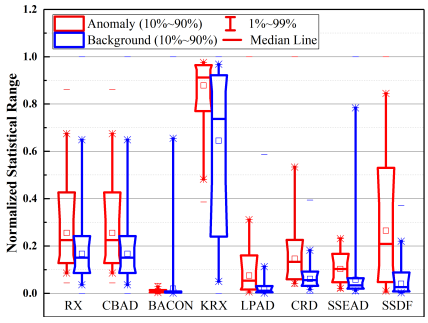


Fig. 7. 2-D color detection maps for the San Diego hyperspectral data set. (a) RX. (b) CBAD. (c) BACON. (d) KRX. (e) LPAD. (f) CRD. (g) SSEAD. (h) SSDF.



(a)



(b)

Fig. 8. Quantitative comparison for the San Diego hyperspectral data set. (a) ROC evaluation. (b) Background-anomaly separation maps.

and BACON have very similar box gaps, while SSEAD gradually increases the gap by the iterative background training process. Compared with the other methods, the SSDF method can alleviate the overlapping of the box and obtains a maximum separation gap for the middle 99% of the two classes.

The 2-D color detection maps of the detection algorithms for the Reno hyperspectral data set are shown in Fig. 11. We can see that the visualization output of KRX is higher than that of the other seven methods, which makes the anomalies difficult to distinguish. We believe that a possible reason for this is that the kernel process of KRX maps the original space into a high-dimensional feature space and increases the spectral difference of the land covers, so the image returns higher detection values after the kernel RX computation. The background values of RX, CBAD, BACON, LPAD, and CRD are fully controlled and most of the anomaly targets are highlighted in the image scene. However, these algorithms obtain higher detection values for some local outliers, which can increase the FAR. In contrast, SSEAD and the proposed SSDF algorithm can suppress the value of the background land cover and avoid detecting local outliers. Since the SSDF algorithm has a better inhibitory effect on the background, its detection map is superior to that of SSEAD. Fig. 12 shows the ROC curves and the background-anomaly separation diagrams for the Reno data set. It can be seen from Fig. 12(a) that when the probability of detection is 0.5, the FAR of the compared algorithms is around 0.01. We believe that the possible reason for the magnified FAR is the complexity of the land-cover types and the potential local outliers that have similar spectral characteristics to the anomalies. As the ROC curve of the proposed SSDF method is clearly superior to those of the other algorithms, this indicates that the axis-parallel subspace process can effectively help eliminate the bands that are unrelated to the anomaly judgment. As for the normalized detection separability maps, it can be seen that RX, CBAD, BACON, LPAD, and CRD have similar gaps between the background and the anomaly boxes. These methods effectively separate the middle 80% statistics of the two classes and suppress the background detection values to a low level. KRX fails to suppress the background, which results in the total energy of the whole image being relatively high. SSEAD and the proposed SSDF method can balance the background suppression

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:17:50 UTC from IEEE Xplore. Restrictions apply.

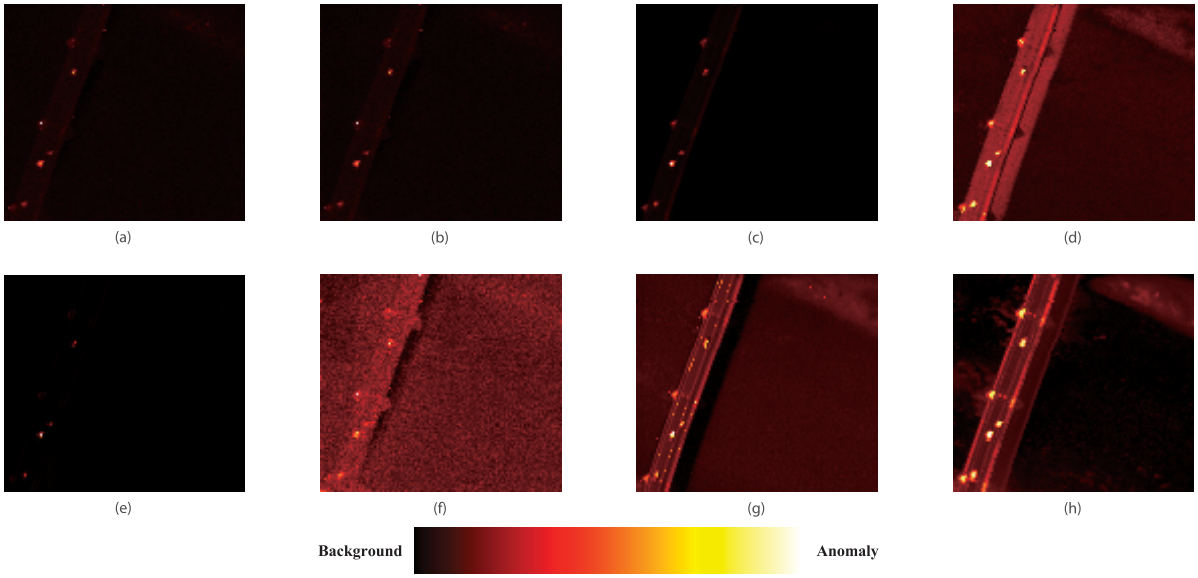


Fig. 9. 2-D color detection maps for the Pavia hyperspectral data set. (a) RX. (b) CBAD. (c) BACON. (d) KRX. (e) LPAD. (f) CRD. (g) SSEAD. (h) SSDF.

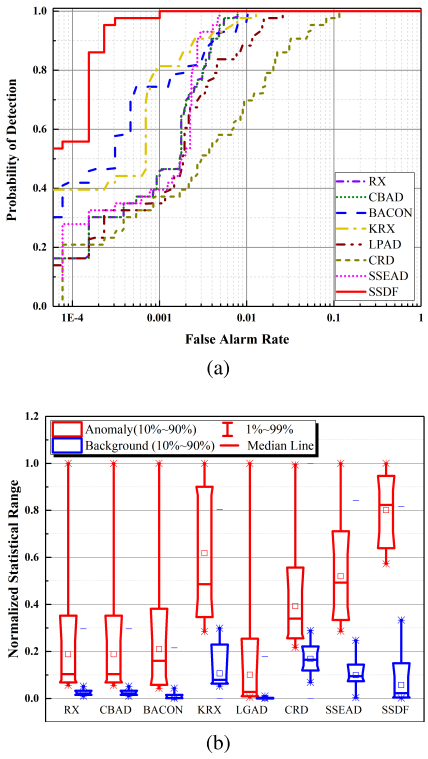


Fig. 10. Quantitative comparison for the Pavia hyperspectral data set. (a) ROC evaluation. (b) Background-anomaly separation maps.

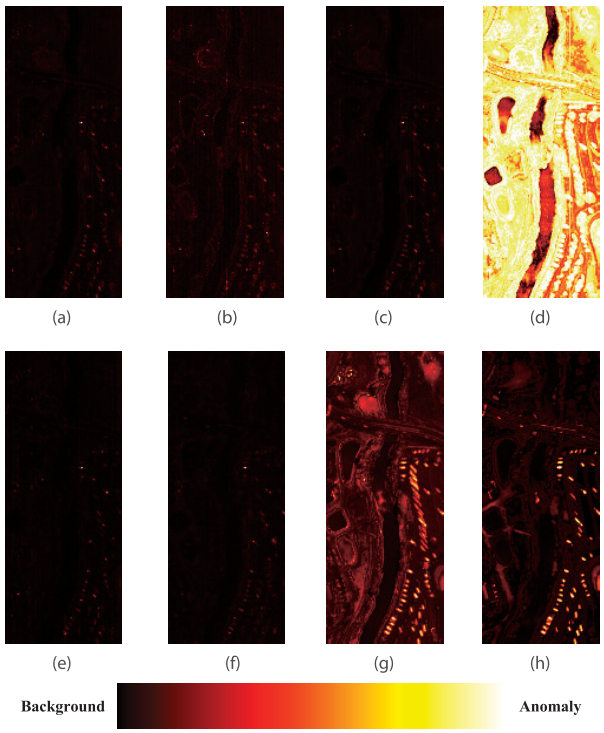
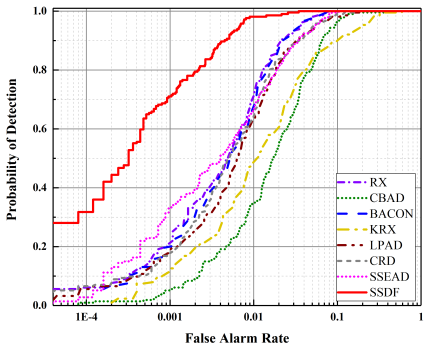


Fig. 11. 2-D color detection maps for the Reno hyperspectral data set. (a) RX. (b) CBAD. (c) BACON. (d) KRX. (e) LPAD. (f) CRD. (g) SSEAD. (h) SSDF.

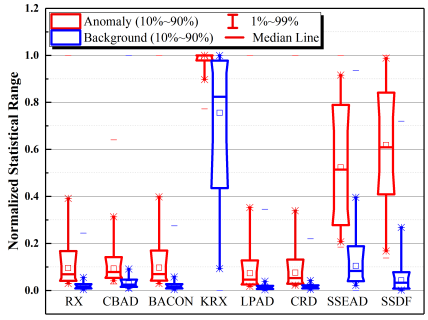
and anomaly enhancement. In comparison, the range of the background detection values is smaller for SSDF, so the separability gap between the two classes is greater. Therefore, the background-anomaly separability map of the proposed SSDF method is better than those of the other methods.

Fig. 13 shows the color detection maps of the compared methods and the proposed SSDF method for the Nuance hyperspectral data set. It can be seen that the proposed SSDF method accurately recognizes the anomaly pixels and achieves good extrusion of the anomalies. RX, CBAD, BACON, and LPAD obtain similar visual detection maps. The total output values of the aforementioned methods are quite low, but they fail to extrude the anomaly pixels. KRX is weak in suppressing the background and noise. Among the different methods, CRD, SSEAD, and the proposed SSDF method obtain better detection performances and successfully highlight the anomalies. Since the background is not fully suppressed by CRD and SSEAD, they perform worse than SSDF. A quantitative

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:17:50 UTC from IEEE Xplore. Restrictions apply.



(a)



(b)

Fig. 12. Quantitative comparison for the Reno hyperspectral data set. (a) ROC evaluation. (b) Background-anomaly separation maps.

comparison of the eight detectors with ROC curves and the normalized background-anomaly separability maps for the nuance data set is shown in Fig. 14. It can be seen from Fig. 14(a) that the DP of KRX is 0 when the FAR is less than 0.002, which indicates that the kernel process cannot improve the detection efficiency but instead magnifies the false alarms. Meanwhile, the detection performances of RX and CRD are inefficient. The ROC curves of SSEAD and the proposed SSDF algorithm are better than those of the other algorithms. When the FAR is less than 0.0004, the DP of the SSDF algorithm is lower than that of SSEAD. However, its DP is higher than that of SSEAD when the FAR is greater than 0.0004. Considering that the ROC curves of the two algorithms intersect, we use the anomaly background separation diagram to compare the detection efficiency. Since the background box of the proposed SSDF method is slightly lower than that of SSEAD, the background-anomaly separation range is better than that of the other algorithms, as depicted in Fig. 14(b).

Finally, a summary of the AUC values together with running times for all of the methods on the four real-world hyperspectral data sets are provided in Table I, and the best AUC performance for each experimental image is highlighted. From the previous demonstration of the experimental results and Table I, we can see that the proposed SSDF method has the best detection efficiency, with the highest AUC values. As for the computational time, our method is the middle of several comparable algorithms.

3) Parameter and Robustness Analysis: In this section, we provide a discussion about the detection performance of the proposed SSDF method, as related to the parameters and the robustness against noises, with the aforementioned data sets. There are three parameters that need to be evaluated: the subsample number n we select as the initial tree elements of each discriminative tree, the number of bands l we use to compute the tree-splitting criterion, and the tree size of N_t the discriminative forest.

First, we analyze the detection accuracy of the SSDF algorithm with regard to the number of tree root nodes, i.e., the number of subsamples. According to the image sizes of the different data sets, the subsample numbers were adjusted to analyze the AUC values of the proposed SSDF algorithm. At this point, the number of subspaces selected for each discriminative tree was set to 20, and the total number of trees in the discriminative forest was set to 1000 by experimental. We know that the reason for the discriminative forest method being effective for anomaly detection is that the subsample strategy can appropriately express the background and anomaly information, using the set of subsamples to simulate and estimate the whole image. When the subsample number of the discriminative tree is too large, more background and noise information will be selected to construct the binary tree, which can cause greater interference and reduced reliability. Therefore, it can be seen from Fig. 15 that the detection efficiency of SSDF reduces if the number of tree elements is too large.

For the San Diego data set, which contains 3600 pixels, we set the subsample number as $n = [400, 2800]$. Through observation, we can see that the detection performance of SSDF first increases and then decreases as the selected subsample number increases. When $n = 2000$, the SSDF algorithm obtains the best performance. For the Pavia data set, which contains 12 960 pixels, we set the subsample number as $n = [500, 5000]$. It can be seen from Fig. 15(b) that when $n = 1000$, the proposed algorithm obtains the best results. For the Reno data set, which contains 25 000 pixels, we set the subsample number as $n = [500, 5000]$. It can be seen from Fig. 15(c) that when $n = 500$, the proposed algorithm obtains the best results. For the Nuance data set, which contains 1 60 000 pixels, we set the subsample number as $n = [2500, 50\,000]$. It can be seen from Fig. 15(d) that when $n = 10\,000$, the proposed algorithm obtains the best results.

We then take a joint evaluation for the number of bands l and the number of trees N_t, and the AUC values of SSDF on four data sets are depicted in Fig. 16(a)–(d), respectively. It can be clearly seen that the detection efficiency of SSDF is not closely related to the tree number N_t, which means that when the subsample number is set to an optimal value, each tree in the discriminative forest models a similar isolating information of training instances, therefore, returns stable anomaly scores at the scoring stage. Considering the time-consuming nature of the discriminative forest ensemble stage, we set the number of trees to 500. On the other hand, it can be seen that the detection result of the SSDF method is greatly influenced by the number of selected bands l. As l growing, the AUC values in four data sets have a tendency of increasing at first and then decreasing. The main reason is that the spectral characteristics of the land covers vary in different image scenes, the number of bands that utilized to distinguish the anomalies from thebackground

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:17:50 UTC from IEEE Xplore. Restrictions apply.

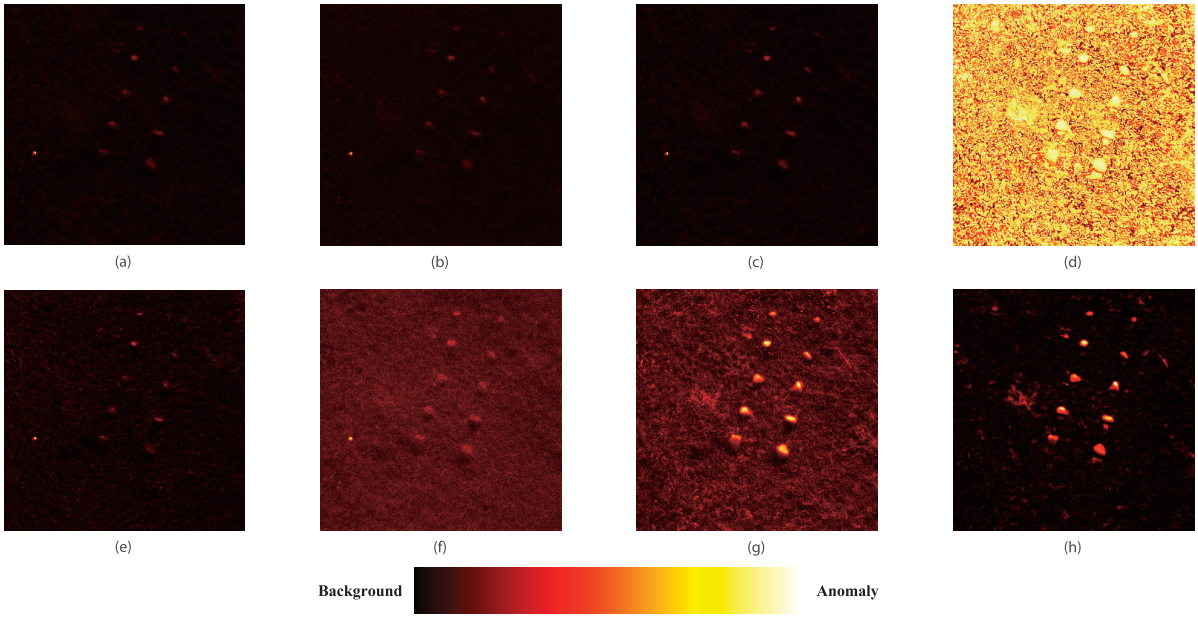
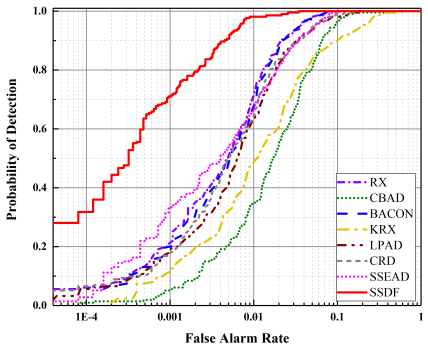
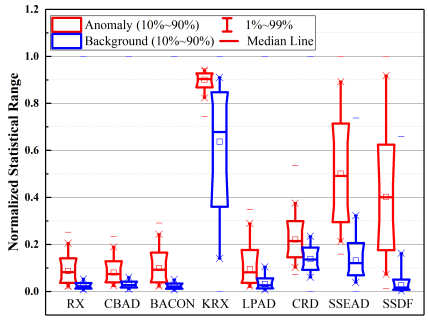


Fig. 13. 2-D color detection maps for the nuance hyperspectral data set. (a) RX. (b) CBAD. (c) BACON. (d) KRX. (e) LPAD. (f) CRD. (g) SSEAD. (h) SSDF.



(a)



(b)

Fig. 14. Quantitative comparison for the nuance hyperspectral data set. (a) ROC evaluation. (b) Background-anomaly separation maps.

TABLE I

AUCs OF THE EXPERIMENTAL RESULTS FOR THE SAN DIEGO, PAVIA, RENO, AND NUANCE DATA SETS

Data set	Algorithm/AUC value/Running time			
San Diego	RX	CBAD	BACON	KRX
	0.8742	0.9449	0.8768	0.7490
	0.6688	0.1060	2.2422	27.5171
	LPAD	CRD	SSEAD	SSDF
	0.9174	0.9607	0.9726	**0.9766**
	0.0830	96.8924	7.7867	6.0021
Pavia	RX	CBAD	BACON	KRX
	0.9878	0.9878	0.9909	0.9915
	0.1324	0.1657	21.9266	64.1041
	LPAD	CRD	SSEAD	SSDF
	0.9758	0.9130	0.9892	**0.9959**
	0.1376	311.7207	40.2794	11.7481
Reno	RX	CBAD	BACON	KRX
	0.9847	0.9519	0.9841	0.9388
	0.6405	1.0082	97.5585	299.0070
	LPAD	CRD	SSEAD	SSDF
	0.9768	0.9787	0.9803	**0.9973**
	0.7453	1279.3192	108.1391	30.5512
Nuance	RX	CBAD	BACON	KRX
	0.9677	0.9518	0.9782	0.9760
	0.6001	0.9219	803.5893	218.9232
	LPAD	CRD	SSEAD	SSDF
	0.9130	0.8822	0.9936	**0.9954**
	0.7167	4359.0396	1852.9866	43.4291

is not the same. For the San Diego data set, the detection performance of SSDF is best when $l = 40$, since this data have totally 189 bands, the optimal axis-parallel subspace for the discriminative tree occupies around 20% of the whole space. For the Pavia data set, the SSDF method obtains the best result when $l = 55$, and the optimal axis-parallel subspace to separate the anomalies accounts for 50% of the whole space. It can be seen from Fig. 15(c) that when $l = 60$, the proposed SSDF method reaches the best result, which illustrates that the optimal proportion of subspace for the Reno data set is 20%. For the Nuance data set, the detection result of SSDF is best when we set $l = 15$, which represents that the subspaces to discriminate abnormal information account for 30% of the total space.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:17:50 UTC from IEEE Xplore. Restrictions apply.

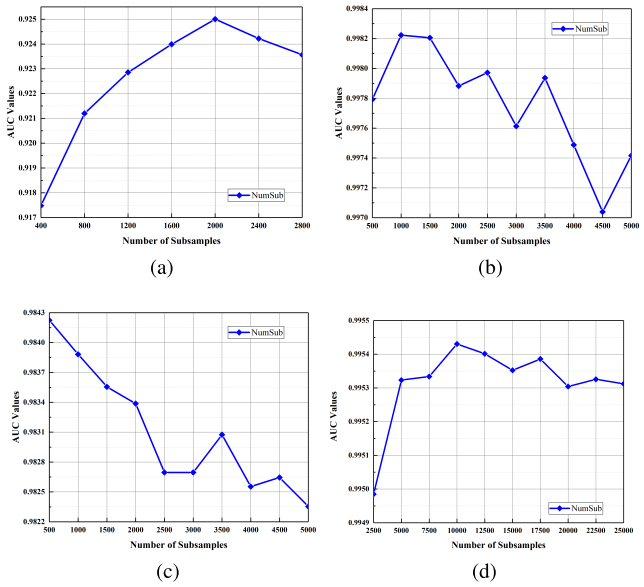


Fig. 15. AUC illustration for the parameter assessment with four hyperspectral data sets. (a) San Diego data set. (b) Pavia data set. (c) Reno data set. (d) Nuance data set.

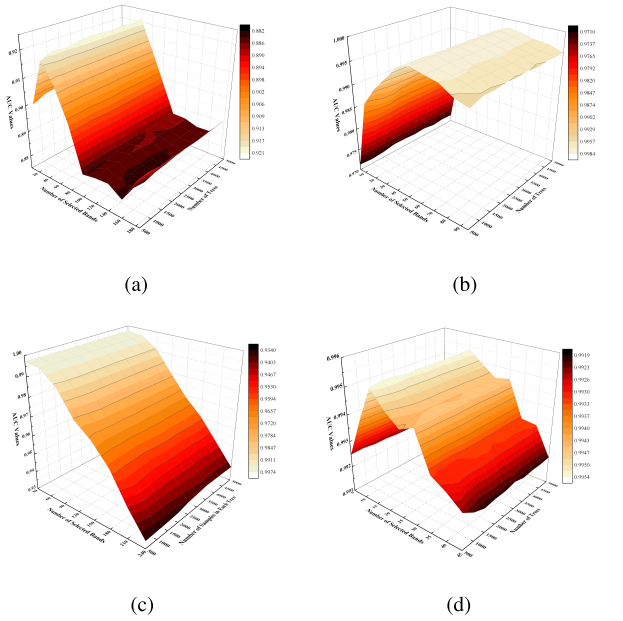


Fig. 16. Performance of SSDF with respect to selected bands and tree size for four hyperspectral data sets. (a) San Diego data set. (b) Pavia data set. (c) Reno data set. (d) Nuance data set.

Considering the tree elements are randomly selected in the *dTree* construction process which may cause fluctuating results, a repeat of 30 times' experiments has been carried out to verify the robustness of the SSDF method. For the San Diego data set, the AUC values under 30 replicates are 0.9759 ± 0.0013. For the Pavia data set, the AUC values are 0.9963 ± 0.0007. For the Reno data set, the AUC values under 30 replicates vary less than $4e-5$. And for the Nuance data set, the AUC values are 0.9952 ± 0.0005. Results prove that the proposed method has strong stability.

Finally, the robustness against noise of the proposed SSDF method is tested in Nuance data set. To conduct our experiments, Gaussian noise with different SNRs are added, and Fig. 17 shows the AUC values of the SSDF method under different SNRs. The SNRs are set from 10 to 50, and the first bar represents AUC value detected from the pure image without noise. It can be seen that as SNR increases, the detection efficiency of SSDF methods improves. When the noise occupies 10% of the image, the AUC value of the SSDF method is only 0.3% different from using the pure image. And when the SNRs are greater than 30, the AUC values become more stable and closer to the original image detection result, which indicates that the proposed method has robustness against noise.

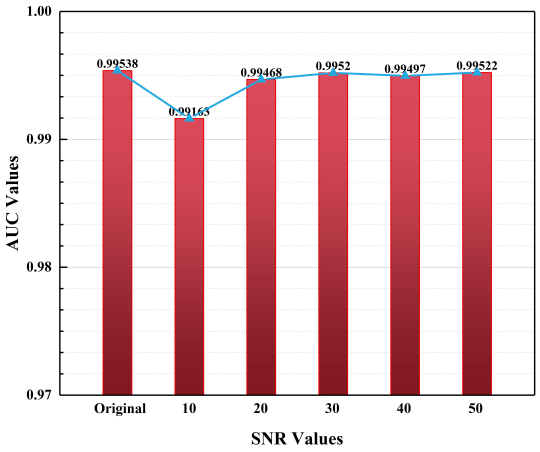


Fig. 17. Robustness performance of SSDF against noise for the nuance data set.

From the aforementioned analyses, it can be concluded that the detection efficiency of the SSDF method is affective by n and l, but not by N_t. This indicates that when the number of subsamples and the number of subspace are set properly, the discriminative tree can reasonably learn the dispersion of the current tree node and separate the anomalies from the background, thus effectively model the overall abnormal information of the data set with partial instances. In addition, the experiments organized on the Nuance data verify that our method has good performances in combating noise.

IV. Conclusion

In this article, with the combination of the axis-parallel subspace selection process and binary tree discrimination, we have proposed the SSDF algorithm to solve the hyperspectral anomaly detection problem. The isolation-based method enables discriminative trees to approximate the overall data distribution by building partial models that concentrate on separating the isolated points from the background pixels. In order to retain the information of the original space and reduce the spectral similarity between bands, an axis-parallel subspace selection process is applied to select those bands that help to detect anomalies. Extensive experiments with four different HSI data sets confirmed that the proposed anomaly detection method outperforms the other state-of-the-art anomaly detection methods. The empirical parameter ranges were also summarized and verified in our experiments, which proved that the proposed SSDF algorithm framework is robust and of practical use.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:17:50 UTC from IEEE Xplore. Restrictions apply.

ACKNOWLEDGMENT

The authors would like to thank the handling editor and the anonymous reviewers for their careful reading and helpful remarks.

REFERENCES

[1] X. Lu, H. Wu, Y. Yuan, P. Yan, and X. Li, "Manifold regularized sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2815–2826, May 2013.

[2] S. Chang, B. Du, L. Zhang, and R. Zhao, "IBRS: An iterative background reconstruction and suppression framework for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 7, pp. 3406–3417, Jul. 2017.

[3] E. Louchard *et al.*, "Derivative analysis of absorption features in hyperspectral remote sensing data of carbonate sediments," *Opt. Express*, vol. 10, no. 26, pp. 1573–1584, Dec. 2002.

[4] R. Merton and J. Huntington, "Early simulation results of the ARIES-1 satellite sensor for multi-temporal vegetation research derived from AVIRIS," in *Proc. 8th JPL Airborne Earth Sci. Workshop*, Pasadena CA, USA, 1999, pp. 299–307.

[5] M. Govender, K. Chetty, and H. Bulcock, "A review of hyperspectral remote sensing and its application in vegetation and water resource studies," *Water Sa*, vol. 33, no. 2, pp. 145–152, Nov. 2011.

[6] P. Goel, S. Prasher, R. Patel, J. Landry, R. Bonnell, and A. Viau, "Classification of hyperspectral data by decision trees and artificial neural networks to identify weed stress and nitrogen status of corn," *Comput. Electron. Agricult.*, vol. 39, no. 2, pp. 67–93, May 2003.

[7] Q. Tian, "Study on oil–gas reservoir detecting methods using hyperspectral remote sensing," *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, vol. 39, pp. 157–162, Aug. 2012.

[8] M. Gålfalk, G. Olofsson, and D. Bastviken, "Approaches for hyperspectral remote flux quantification and visualization of GHGs in the environment," *Remote Sens. Environ.*, vol. 191, pp. 81–94, Mar. 2017.

[9] D. Manolakis, D. Marden, and G. A. Shaw, "Hyperspectral image processing for automatic target detection applications," *Lincoln Lab. J.*, vol. 14, no. 1, pp. 79–116, 2003.

[10] X. Lu, Y. Wang, and Y. Yuan, "Graph–regularized low–rank representation for destriping of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 7, pp. 4009–4018, Jul. 2013.

[11] S. Yang and Z. Shi, "Hyperspectral image target detection improvement based on total variation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2249–2258, May 2016.

[12] Z. Wang, R. Zhu, K. Fukui, and J.-H. Xue, "Matched shrunken cone detector (MSCD): Bayesian derivations and case studies for hyperspectral target detection," *IEEE Trans. Image Process.*, vol. 26, no. 11, pp. 5447–5461, Nov. 2017.

[13] D. Manolakis, C. Siracusa, and G. Shaw, "Hyperspectral subpixel target detection using the linear mixing model," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 7, pp. 1392–1409, Jul. 2001.

[14] Z. Zou and Z. Shi, "Random access memories: A new paradigm for target detection in high resolution aerial remote sensing images," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1100–1111, Mar. 2018.

[15] Y. Zhang, B. Du, L. Zhang, and T. Liu, "Joint sparse representation and multitask learning for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 894–906, Feb. 2017.

[16] J. E. Fowler and Q. Du, "Anomaly detection and reconstruction from random projections," *IEEE Trans. Image Process.*, vol. 21, no. 1, pp. 184–195, Jan. 2012.

[17] S. Chang, B. Du, and L. Zhang, "BASO: A background–anomaly component projection and separation optimized filter for anomaly detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 3747–3761, Jul. 2018.

[18] S. Matteoli, M. Diani, and G. Corsini, "A tutorial overview of anomaly detection in hyperspectral images," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 25, no. 7, pp. 5–28, Jul. 2010.

[19] S. Matteoli, T. Veracini, M. Diani, and G. Corsini, "Background density nonparametric estimation with data–adaptive bandwidths for the detection of anomalies in multi–hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 163–167, Jan. 2014.

[20] Y. Yuan, D. Ma, and Q. Wang, "Hyperspectral anomaly detection by graph pixel selection," *IEEE Trans. Cybern.*, vol. 46, no. 12, pp. 3123–3134, Dec. 2016.

[21] B. Du and L. Zhang, "A discriminative metric learning based anomaly detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 6844–6857, Nov. 2014.

[22] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Joint reconstruction and anomaly detection from compressive hyperspectral images using Mahalanobis distance–regularized tensor RPCA," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2919–2930, May 2018.

[23] R. Zhao, B. Du, L. Zhang, and L. Zhang, "A robust background regression based score estimation algorithm for hyperspectral anomaly detection," *ISPRS J. Photogram. Remote Sens.*, vol. 122, pp. 126–144, Dec. 2016.

[24] R. Zhao, B. Du, L. Zhang, and L. Zhang, "Beyond background feature extraction: An anomaly detection algorithm inspired by slowly varying signal analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1757–1774, Mar. 2016.

[25] I. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[26] M. Carlotto, "A cluster-based approach for detecting man-made objects and changes in imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 374–387, Feb. 2005.

[27] N. Billor, A. S. Hadi, and P. F. Velleman, "BACON: Blocked adaptive computationally efficient outlier nominators," *Comput. Statist. Data Anal.*, vol. 34, no. 3, pp. 279–298, Sep. 2000.

[28] B. Du and L. Zhang, "Random-selection-based anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 5, pp. 1578–1589, May 2011.

[29] D. W. J. Stein, S. G. Beaven, L. E. Hoff, E. M. Winter, A. P. Schaum, and A. D. Stocker, "Anomaly detection from hyperspectral imagery," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 58–69, Jan. 2002.

[30] H. Kwon and N. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[31] B. Schölkopf, A. Smola, and K.-R. Müller, "Kernel principal component analysis," in *Proc. Int. Conf. Artif. Neural Netw.*, 1997, pp. 583–588.

[32] R. Zhao, B. Du, and L. Zhang, "A robust nonlinear hyperspectral anomaly detection approach," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 4, pp. 1227–1234, Apr. 2014.

[33] A. Banerjee, P. Burlina, and C. Diehl, "A support vector method for anomaly detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 8, pp. 2282–2291, Aug. 2006.

[34] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Top. Appl. Earth Observat. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.

[35] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[36] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low–rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[37] X. Zhang, G. Wen, and W. Dai, "A tensor decomposition–based anomaly detection algorithm for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 5801–5820, Oct. 2016.

[38] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, "LOF: Identifying density-based local outliers," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, 2000, vol. 29, no. 2, pp. 93–104.

[39] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, and R. C. Williamson, "Estimating the support of a high–dimensional distribution," *Neural Comput.*, vol. 13, no. 7, pp. 1443–1471, Jul. 2001.

[40] S. Ramaswamy, R. Rastogi, and K. Shim, "Efficient algorithms for mining outliers from large data sets," *SIGMOD Rec.*, vol. 29, no. 2, pp. 427–438, Jun. 2000.

[41] C. C. Aggarwal, *High-Dimensional Outlier Detection: The Subspace Method*. Cham, Switzerland: Springer, 2013, pp. 135–167.

[42] H.-P. Kriegel, P. Kröger, E. Schubert, and A. Zimek, "Outlier detection in axis-parallel subspaces of high dimensional data," in *Proc. Pacific–Asia Conf. Knowl. Discovery Data Mining*, 2009, pp. 831–838.

[43] C. C. Aggarwal and P. S. Yu, "Outlier detection for high dimensional data," *SIGMOD Rec.*, vol. 30, no. 2, pp. 37–46, Jun. 2001.

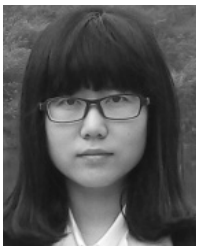
[44] F. T. Liu, K. M. Ting, and Z. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. Data Mining*s, Dec. 2008, pp. 413–422.

[45] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation-based anomaly detection," *ACM Trans. Knowl. Discovery Data*, vol. 6, no. 1, p. 3, 2012.

[46] D. Zhang, N. Li, Z.-H. Zhou, C. Chen, L. Sun, and S. Li, "IBAT: Detecting anomalous taxi trajectories from GPS traces," in *Proc. 13th Int. Conf. Ubiquitous Comput.*, 2011, pp. 99–108.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:17:50 UTC from IEEE Xplore. Restrictions apply.

[47] X. Liu *et al.*, "Semi-supervised node splitting for random forest construction," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 492–499.
[48] M. Pal, "Random forest classifier for remote sensing classification," *Int. J. Remote Sens.*, vol. 26, no. 1, pp. 217–222, Jan. 2005.
[49] D. R. H. P. Pattern, *Classification and Scene Analysis*. Hoboken, NJ, USA: Wiley, 1973.
[50] Y. Liu, G. Gao, and Y. Gu, "Tensor matched subspace detector for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 1967–1974, Apr. 2017.
[51] T. Zhou, D. Tao, and X. Wu, "Manifold elastic net: A unified framework for sparse dimension reduction," *Data Min Knowl Disc*, vol. 22, no. 3, pp. 340–371, May 2011.
[52] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low–rank and sparse matrix decomposition–based mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.
[53] J. Kerekes, "Receiver operating characteristic curve confidence intervals and regions," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 2, pp. 251–255, Apr. 2008.
[54] Y. Niu and B. Wang, "Hyperspectral anomaly detection based on low–rank representation and learned dictionary," *Remote Sens.*, vol. 8, no. 4, p. 289, Mar. 2016.
[55] D. Manolakis and G. Shaw, "Detection algorithms for hyperspectral imaging applications," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 29–43, Jan. 2002.
[56] Y. Zhang, L. Zhang, B. Du, and S. Wang, "A nonlinear sparse representation–based binary hypothesis model for hyperspectral target detection," *IEEE J. Sel. Top. Appl. Earth Observat. Remote Sens.*, vol. 8, no. 6, pp. 2513–2522, Jun. 2015.
[57] Z. Li, L. Wang, and S. Zheng, "Applied low dimension linear manifold in hyperspectral imagery anomaly detection," *Proc. SPIE*, vol. 9142, Feb. 2014, Art. no. 91421P.
[58] R. Zhao, B. Du, and L. Zhang, "Hyperspectral anomaly detection via a sparsity score estimation framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3208–3222, Jun. 2017.



Shizhen Chang received the B.S. degree in surveying and mapping engineering from Wuhan University, Wuhan, China, in 2016, where she is currently pursuing the Ph.D. degree with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing.

Her research interests include hyperspectral image processing, target detection, and machine learning.



Bo Du (Senior Member, IEEE) received the B.S. and Ph.D. degrees in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote sensing, Wuhan University, Wuhan, China, in 2010.

He is currently a Professor with the National Engineering Research Center for Multimedia Software, Institute of Artificial Intelligence and School of Computer Science, Wuhan University. He has authored or coauthored more than 60 research articles in the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEM (TNNLS), the IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), the IEEE TRANSACTIONS ON MULTIMEDIA (TMM), the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), the IEEE JOURNAL OF SELECTED TOPICS IN EARTH OBSERVATIONS AND APPLIED REMOTE SENSING (JSTARS), and the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL), etc., 13 of which are ESI hot articles or highly cited articles. His major research interests include pattern recognition, hyperspectral image processing, machine learning and signal processing.

Dr. Du received the Highly Cited Researcher 2019 Award from the Web of Science Group, the Distinguished Paper Award from IJCAI 2018, the Best Paper Award of the IEEE Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing (WHISPERS) 2018, and the Champion Award of the IEEE Data Fusion Contest 2018. He received the Best Reviewer Awards from the IEEE GRSS for his service to the IEEE JSTARS in 2011 and ACM Rising Star Awards for his academic progress in 2015. He serves as an Associate Editor for *Pattern Recognition and Neurocomputing*, as Senior PC for IJCAI/AAAI/KDD, and as the Area Chair for the Indian Council of Philosophical Research (ICPR) and the International Joint Conference on Neural Networks (IJCNN). He was the Session Chair for both the International Geoscience and Remote Sensing Symposium (IGARSS) 2018/2016 and the 4th IEEE GRSS WHISPERS. He also serves as a Reviewer for 20 Science Citation Index (SCI) magazines, including the IEEE TGRS, IEEE TIP, IEEE JSTARS, and IEEE GRSL.



Liangpei Zhang (Fellow, IEEE) received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 1982, the M.S. degree in optics from the Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, in 1988, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1998.

He is currently a Chang-Jiang Scholar Chair Professor appointed by the Ministry of Education of China with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote sensing (LIESMARS), Wuhan University. He was a Principal Scientist for the China State Key Basic Research Project from 2011 to 2016 appointed by the Ministry of National Science and Technology of China to lead the Remote Sensing Program in China. He has authored or coauthored more than 700 research articles and 5 books. He is a highly cited author in the Institute for Scientific Information (ISI). He holds 30 patents. His research interests include hyperspectral remote sensing, high-resolution remote sensing, image processing, and artificial intelligence.

Dr. Zhang is a fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2010 Best Paper Boeing Award, the 2013 Best Paper ERDAS Award from the American Society of Photogrammetry and Remote Sensing (ASPRS), and the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). His research teams have won the top three prizes of the IEEE GRSS 2014 Data Fusion Contest, and his students have been selected as the winners or finalists of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS) Student Paper Contest in recent years. He also serves as an Associate Editor or Editor for more than ten international journals. He is currently serving as an Associate Editor for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He is the Founding Chair of the IEEE Geoscience and Remote Sensing Society (GRSS) Wuhan Chapter.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:17:50 UTC from IEEE Xplore. Restrictions apply.